

EXECUTIVE SUMMARY

V-GUARD INDUSTRIES LTD – BIG IDEA TECH DESIGN COMPETITION 2026

(The first page of the executive summary must adhere to the format given below)

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Google Drive Video Link

Please paste the google drive link of your team's video pitch (40 seconds) below.

<https://drive.google.com/file/d/1eBcNDpe4yKDJRv7YKbVwHbuWcS2VZ9WY/view?usp=sharing>

(The first page of the executive summary must adhere to the format given below and must not exceed 500 words)

Problem Statement (What & Why)

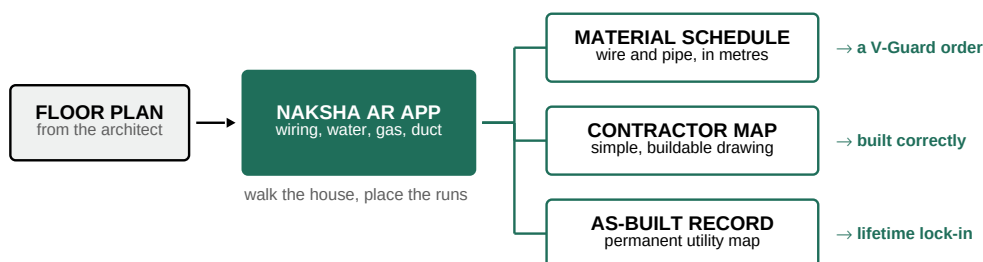
When a family builds a house, they do not know where anything needs to go. The wiring, the water pipes, the gas line, the chimney duct. Even with the architect's drawings it stays confusing, because a flat plan cannot tell you how a room will feel, or where you will actually want a socket, a geyser or a chimney.

So these decisions get made on site, quickly, by whoever happens to be there. Runs are improvised, material is mis-ordered, and nothing is written down. Roughly 42% of building fires in India are attributed to electrical short circuits, and with no record, later renovation becomes guesswork.

For V-Guard, that same moment is a missed commercial opportunity. Every one of those decisions settles which brand of wire, water heater, chimney, RO unit and gas stove ends up in the house. V-Guard sells all of them, and is present for none of it.

Brief Description of the Idea/Solution

NAKSHA is an augmented reality app from V-Guard. The owner walks through the under-construction house holding a phone and places what they want where they want it: sockets and switches, the water heater, the water line, the gas point, the chimney duct. They see it at full scale on the actual wall, before anything is chased or plastered.



The owner designs in AR. The app produces a material order, a drawing the trades can build from, and a permanent record.

The app is a marketing instrument as much as a design tool. At the exact moment the owner is deciding where the water heater goes, V-Guard is the brand in their hand, showing the V-Guard model that fits the space. The app is free. What it sells is wire, water heaters, chimneys, RO systems and gas stoves. V-Guard offers it as a service, and the service sells the products.

Technology Proposed

Augmented reality layer: ARCore and ARKit plane detection with world tracking, so placed runs and appliances stay fixed to the wall as the user moves. Phone based, nothing to carry on site. **Plan understanding:** an architect's plan can optionally be uploaded and segmented into rooms to give real dimensions. **Rule engine:** the national wiring and earthing codes encoded as explicit, auditable constraints, so what the owner draws is corrected to code rather than merely recorded. **Routing and estimation:** runs are routed to minimise length and voltage drop, then converted into wire by size, pipe and duct lengths, boxes and switchgear, mapped onto V-Guard product codes. **Output for trades:** a dimensioned drawing, because a contractor will not use an app. **As-built capture:** each run is confirmed before plastering.

(From the second page: remaining content that supports and justifies the Executive Summary)

Why V-Guard, and not a startup

This is the argument the whole proposal rests on. An independent app that helps you plan your home services has no way to make money except by charging for the software, which nobody in this market will pay for.

V-Guard sells products in every category the app touches: wires and cables, water heaters, chimneys, RO purifiers and gas stoves. That makes the app a sales channel rather than a product. It can be given away free, forever, because the revenue is downstream. **No competitor spans all four utilities, so no competitor can afford to give this away.**

Novelty and Innovation

The components exist and the proposal is stronger for saying so. Automated electrical takeoff is a mature category: tools such as drawer.ai accept PDF drawings and return quantities and wire lengths with marked-up routing, and Kreo, Trimble Accubid and PlanSwift serve the same market. Work presented at ISARC 2026 automates circuit assignment in a CAD and BIM workflow. Robotic site layout is solved too, with Dusty Robotics having printed coordinated models onto over 300 million square feet of slab.

All of it serves Western commercial contractors bidding large projects. Our contribution is positional:

- **The home owner is the user**, not an estimator. Existing tools ask for a finished design; NAKSHA lets a non-technical person create one by walking around and pointing.
- **All four utilities in one pass**, which mirrors how a house is actually built and which only V-Guard's portfolio can monetise.
- **Indian codes as the rule set**. None of the existing tools encode them.
- **The as-built record**, which no takeoff tool produces, because its users leave the site and never return.

The channel is already built

Every major Indian wire brand has already digitised its electrician relationship. RR Kabel has RR Connect, Finolex has Samruddhi, Havells has an electrician app, and Polycab's Experts platform reaches around 2.5 lakh electricians and retailers, described in its FY26 annual report as a grassroots influencer channel.

Every one of them is a loyalty scheme. Scan a code, collect points, redeem a gift. Nobody has put a tool in that channel that does any actual work. The electrician does not want points. He wants the load calculation done, the sizes decided, the material list totalled, and a defensible answer when the owner asks why it costs what it costs. Give him that, and he becomes the distribution.

End Consumer and Business Value

For the owner: the ability to see and decide the house before it is sealed, an itemised material list that makes overcharging visible, a home wired to code rather than to habit, and a permanent map of every service run in the building.

For V-Guard, four things:

- **It is a marketing channel that sells at the moment of decision.** Every appliance the owner places in AR is a V-Guard product shown in their own room, at the point where the choice is actually made.
- **Every design becomes an order.** The output is a material schedule in metres and units, mapped to V-Guard codes. Specifying material is a stronger position than advertising to whoever walks into a

shop. This is the Asian Paints move, which used Beautiful Homes services to escape commodity paint competition, and which Berger followed with Express Painting.

- **It captures the decision years earlier.** Services are specified during construction, before the owner has any brand opinion, and they are never re-shopped. You cannot rewire or re-plumb a finished house.
- **It is a margin story, not a volume story.** FY25-26 revenue grew 7.0% to Rs 5,966 crore while PAT fell 1.7% to Rs 308 crore. Commodity wire cannot fix that; specified systems can.

Delivery and scale

The app scales at software cost and needs no field organisation, which is the whole point of choosing AR over any equipment-based approach. Optional paid layers sit on top: a trained V-Guard marking partner in metro cities, and a robotic layout flagship deployed as a showcase. Treating the flagship as a marketing instrument rather than a revenue line is the honest position, and it is exactly how paint companies launched their service brands.

Risks and honest limitations

- **AR accuracy on a live site is the technical risk.** Drift and poor lighting are real, so the app should promise dimensioned guidance and reference marks rather than millimetre placement, and always emit a measured drawing as the authoritative output.
- **A contractor will not use an app.** This is why the deliverable to the trades is a printed drawing and a material list, not a phone screen.
- **Liability.** A licensed contractor is legally responsible for the installation, so the tool must be a decision aid he signs off, not the designer of record. This needs legal structuring before launch.
- **Adoption, not technology, is the real risk.** The app must visibly save time on load calculation and estimating from day one, or it will be ignored by the trades who decide what gets bought.

What we would build next

- A working AR prototype on a phone: place sockets, a water heater and a pipe run in a real room, and export a dimensioned drawing plus a material list.
- The code rule engine, with every generated decision traceable to a clause.
- An optimisation study of material length against distribution board placement across several house typologies, quantifying the saving a good layout produces over an improvised one.
- An AR registration test on a real wall, reporting measured error honestly rather than asserting it.